4.5 The substitution Rule.

$\int_a^b f(x)\,dx$ (Area)

- limit definition.
- Fundamental thm Calculus P2. ⟶ We need an antiderivative for $f(x)$.

We want

$$\int 2x\sqrt{1+x^2}\,dx$$

To find this integral we use the problem-solving strategy of introducing something extra. Here the "something extra" is a new variable.

Solution: Let $u = 1+x^2$. $du = (0+2x)dx = 2x\,dx$.

$$\int \underbrace{2x\sqrt{1+x^2}}\,dx = \int \sqrt{u}\,du = \int u^{1/2}\,du = \frac{2}{3}u^{3/2} + C$$

$$= \frac{2}{3}(1+x^2)^{3/2} + C.$$

$$\frac{d}{dx}\left\{\frac{2}{3}(1+x^2)^{3/2}+C\right\} = \frac{2}{3}\cdot\frac{3}{2}(1+x^2)^{3/2-1}\cdot\frac{d}{dx}(1+x^2)$$

$$= (1+x^2)^{1/2}(2x)$$

$$= 2x\sqrt{1+x^2}.$$

In general, the method works whenever we have an integral that can be write in the form.

$$\int f(g(x))\cdot g'(x)\,dx \qquad \text{If } F' = f \text{ then}$$

$$\int F'(g(x))g'(x)\,dx = F(g(x)) + C$$

If we make the "change variable" or "substitution" $u = g(x)$ then

$$\int F'(g(x))g'(x)\,dx = \int F'(u)\,du \quad \text{where } F' = f$$
$$= \int f(u)\,du.$$

Thm. If $u=g(x)$ is a differentiable function whose range is an interval and f is continuous on I then.

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du.$$

Example. Find $\int x^3 \cos(x^4+2)\,dx$.

Sln. $f(x) = \cos(x^4+2) = h(g(x))$ where $h(x) = \cos x$. $g(x) \overset{?}{=} x^4+2$

b/c $h(g(x)) = h(x^4+2) = \cos(x^4+2)$.

$g'(x) = 4x^3$. Let $u = x^4+2$ $du = 4x^3dx \Rightarrow x^3dx = \frac{du}{4}$.

$$\int x^3\cos(x^4+2)\,dx = \int \cos(u)\frac{du}{4} = \frac{1}{4}\int \cos u\,du$$

$$= \frac{1}{4}\sin u + C.$$

$$= \frac{1}{4}\sin(x^4+2) + C.$$

Problem: Evaluate $\int \sqrt{2x+1}\,dx$.

Sln? $\int \sqrt{2x+1}\,dx = \int (2x+1)^{1/2}\,dx = \frac{(2x+1)^{1/2+1}}{1/2+1} + C = \frac{2}{3}(2x+1)^{3/2} + C.$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C.$$

$$\frac{d}{dx}\left\{\frac{2}{3}(2x+1)^{3/2}\right\} = \frac{2}{3}\cdot\frac{3}{2}\cdot(2x+1)^{3/2-1}\cdot\frac{d}{dx}(2x+1)$$

$$= \sqrt{2x+1}\cdot(2)$$

$$= 2\sqrt{2x+1}$$

Example. Find $\int \frac{x}{\sqrt{1-4x^2}}\,dx$.

Sln:

$$\int \frac{x}{\sqrt{1-4x^2}}\,dx = \int x\cdot\frac{1}{\sqrt{1-4x^2}}\,dx$$

Let $f(x) = \frac{1}{\sqrt{1-4x^2}} = h(g(x))$ where $h(x) = \frac{1}{\sqrt{x}}$

$g(x) = 1-4x^2 \Rightarrow g'(x) = -8x$.

Let $u = 1-4x^2$ $du = -8x\,dx$ then $x\,dx = -\frac{1}{8}du$.

$$\int \frac{x}{\sqrt{1-4x^2}}dx = \int \frac{1}{\sqrt{u}}\left(-\frac{du}{8}\right) = -\frac{1}{8}\int \frac{1}{\sqrt{u}}du \quad ; \quad x^{-n} = \frac{1}{x^n}$$

$$= -\frac{1}{8}\int u^{-1/2}du$$

$$= -\frac{1}{8}\,\frac{u^{1/2}}{1/2} + C$$

$$= -\frac{1}{4}\sqrt{u} + C$$

$$= -\frac{1}{4}\sqrt{1-4x^2} + C.$$

Example. Evaluate $\int \cos(5x)dx$.

Sln: $\int \cos(5x)dx = \int 1 \cdot \cos(5x)\,dx.$

Let $f(x) = \cos(5x) = h(g(x))$ where $h(x) = \cos x$ $g(x) = 5x$.

$g'(x) = 5$.

Let $u = 5x$ $du = 5dx \iff \frac{du}{5} = dx$.

By subst.

$$\int \cos(5x)dx = \int \cos u \,\frac{du}{5} = \frac{1}{5}\int \cos u\,du. = \frac{1}{5}\sin u + C.$$

$$= \frac{1}{5}\sin(5x) + C.$$

Example. Find $\int \sqrt{1+x^2}\,x^5\,dx$.

Sln. $f(x) = \sqrt{1+x^2} = h(g(x))$ where $h(x) = \sqrt{x}$ $g(x) = 1+x^2$.

$g'(x) = 2x$.

Let $u = 1+x^2$ $du = 2x\,dx \Rightarrow x\,dx = \frac{1}{2}du$.

$$\int \sqrt{1+x^2}\,x^5\,dx = \int \sqrt{1+x^2}\,x^4\,(x\,dx)$$

$$= \int \sqrt{u}\,x^4\,\frac{1}{2}du.$$

We need to find x^4 in terms of u. We know $u = 1+x^2 \Rightarrow x^2 = 1-u$.

then $x^4 = (1-u)^2$.

$$\int \sqrt{1+x^2}\; x^4 (x\,dx) = \int \sqrt{u}\,(1-u)^2 \frac{du}{2}$$

$$= \frac{1}{2}\int \sqrt{u}\,(1-2u+u^2)\,du$$

$$= \frac{1}{2}\int (u^{1/2} - 2u^{3/2} + u^{5/2})\,du$$

$$= \frac{1}{2}\left\{\frac{2}{3}u^{3/2} - 2\cdot\frac{2}{5}u^{5/2} + \frac{2}{7}u^{7/2}\right\} + C$$

$$= \frac{1}{3}u^{3/2} - \frac{2}{5}u^{5/2} + \frac{1}{7}u^{7/2} + C$$

$$= \frac{1}{3}(1+x^2)^{3/2} - \frac{2}{5}(1+x^2)^{5/2} + \frac{1}{7}(1+x^2)^{7/2} + C$$

Substitution : Definite integrals

Evaluate $\int_0^4 \sqrt{2x+1}\,dx$

Soln 1: **Step 1:** (find an antiderivative).

$\int \sqrt{2x+1}\,dx$ let $u = 2x+1$ $du = 2\,dx$ $dx = \frac{1}{2}du$

$$= \frac{1}{2}\int \sqrt{u}\,du = \frac{1}{2}\cdot\frac{2}{3}u^{3/2} = \frac{1}{3}(2x+1)^{3/2}$$

Step 2: (Apply FTC P2):

$$\int_0^4 \sqrt{2x+1}\,dx = \frac{1}{3}(2x+1)^{3/2}\Big|_0^4 = \frac{1}{3}(8+1)^{3/2} - \frac{1}{3} = \frac{1}{3}(9)^{3/2} - \frac{1}{3}$$

$$= \frac{1}{3}(27-1) = \frac{26}{3}.$$

Soln 2.

$$\int_0^4 \sqrt{2x+1}\,dx = \int_1^9 \sqrt{u}\,\frac{du}{2}$$

$$= \frac{1}{2}\int_1^9 \sqrt{u}\,du$$

$$= \frac{1}{2}\cdot\frac{2}{3}(u)^{3/2}\Big|_1^9$$

$$= \frac{1}{3}\{9^{3/2} - 1^{3/2}\} = \frac{1}{3}\{27-1\} = \frac{26}{3}.$$

$u = 2x+1$

$du = 2\,dx \Rightarrow dx = \frac{1}{2}du$

if $x=0$ then $u=1$

if $x=4$ then $u=9$

Thm. (The substitution rule for Definite integrals) If g' is continuous on $[a,b]$ and f is continuous on the range of $u=g(x)$ then

$$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

Example. Evaluate $\int_1^2 \frac{dx}{(3-5x)^2}$.

Sln. Let $u = 3-5x$ $\quad du = -5\,dx$ $\quad dx = -\frac{1}{5}du$.

if $x=1$ then $u=-2$.

if $x=2$ then $u=-7$.

$$\int_1^2 \frac{dx}{(3-5x)^2} = -\frac{1}{5}\int_{-2}^{-7} \frac{1}{u^2}\,du = -\frac{1}{5}\left(-\int_{-7}^{-2} u^{-2}\,du\right)$$

$$= \frac{1}{5}\int_{-7}^{-2} u^{-2}\,du$$

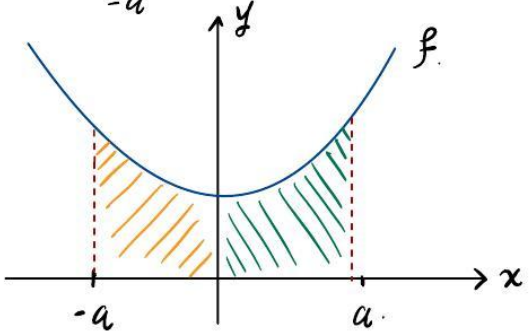
$$= -\frac{1}{5}\left.\frac{1}{u}\right|_{-7}^{-2} = -\frac{1}{5}\left(-\frac{1}{2}+\frac{1}{7}\right) = \frac{1}{14}.$$

Symmetry: Recall that a function f is even if $f(-x)=f(x)$

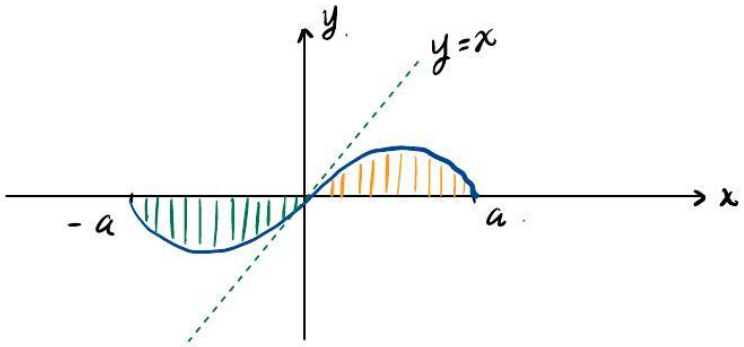
f is odd if $f(-x)=-f(x)$.

Thm: (Integrals of Symmetric functions) Suppose that f is continuous on $[-a,a]$.

a). If f is even then $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$.



b) If f is odd then $\int_{-a}^{a} f(x)\,dx = 0$.



Example. Because $f(x)=x^6+1$ satisfies $f(-x)=(-x)^6+1=x^6+1=f(x)$ it is an even function

$$\int_{-2}^{2}(x^6+1)\,dx = 2\int_{0}^{2}(x^6+1)\,dx = 2\left\{ \frac{x^7}{7}+x \Big|_0^2 \right\}.$$

$$= 2\left\{ \frac{2^7}{7}+2 \right\} = \frac{284}{7}.$$

Problem for you! Compute $\displaystyle\int_{-1}^{1}\frac{\tan x}{1+x^2+x^4}\,dx.$